**Faculty of Computers and Data Science**

**Intelligent Systems Department**

**NLP Project**

# Arabic Sentiment Analysis

# *Team Members*

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# 1 Introduction
Sentiment analysis is like understanding how people feel. It's a way for computers to figure out if a piece of writing, like a review or a social media post, expresses positive, negative, or neutral emotions.

Imagine you're reading a message or a comment from someone. Sentiment analysis is like having a tool that helps you quickly know if the person is happy, sad, excited, angry, or just sharing information without any strong emotion.

Computers use special techniques to study the words, phrases, or even emojis in a text to guess the emotions behind them. This helps companies understand how their customers feel about their products or services by analyzing online reviews or social media comments. It can also assist in tracking public opinions about certain topics or events.

## 1.1 Project Idea
Create a comprehensive system that performs sentiment analysis and topic modeling on Arabic text data. This project involves processing, analyzing, and interpreting textual data in Arabic to extract sentiment and understand prevalent topics within the content.

## 1.2 Data Used
HTL.csv

### 1.2.1 Source
Github

### 1.2.2 Data Description
- Dataset of Hotel Reviews scrapped from TripAdvisor.com
- Consist of 15572 reviews

## 1.3 Objectives

- Develop a Bidirectional LSTM model capable of performing sentiment analysis on Arabic text data.
- Train the model using the collected dataset to classify text sentiment as positive, negative.
- Evaluate and optimize the model's accuracy and performance metrics for reliable sentiment predictions.

# 2 Data Preprocessing and text cleaning

1. Arabic Diacritics Removal:

Diacritics in Arabic texts, such as Tashkeel , Fatha , Damma , Kasra , etc., are removed from the text.

2. Text Cleaning:

Text undergoes a cleaning process where various transformations occur:

- Elongated characters (repeated letters) are reduced to their original form (e.g., "جميل" from "جميييييييييييييل").
- Non-alphanumeric characters, punctuation, and symbols are removed.
- English letters, digits, and special characters are eliminated as they don't contribute to the sentiment or topic analysis of Arabic text.
- Unnecessary whitespace, tabs, newline characters, and carriage returns are replaced or removed.

3. Normalization of Arabic Characters:

Specific Arabic characters with alternate forms or different representations are standardized to their most common form. For instance, different variations of the same character (like 'أ', 'إ', 'آ' to 'ا') are unified to simplify the text.

4. Balancing Positive and Negative Samples:

To ensure a balanced dataset, an equal number of positive and negative samples are selected to avoid bias in sentiment analysis.

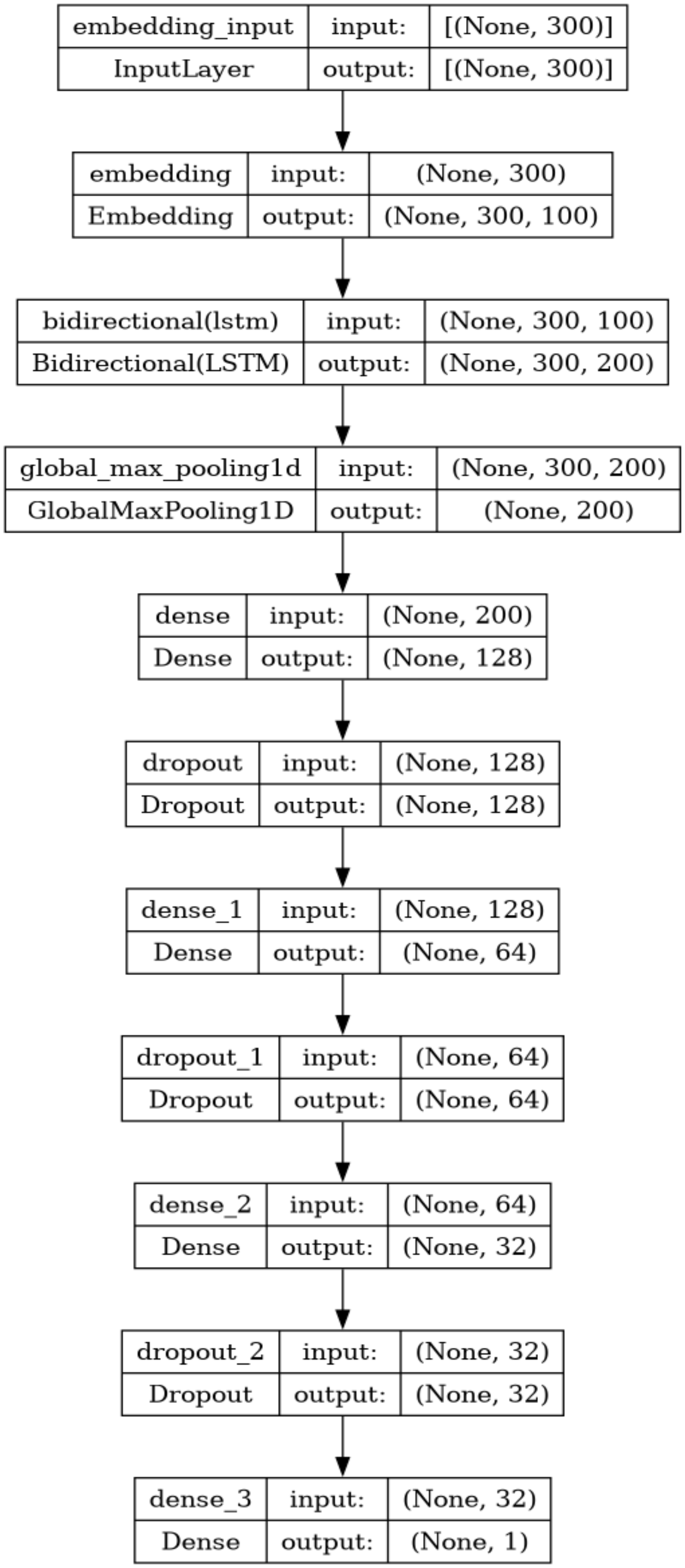
5. Tokenization and Padding:

The text is converted into sequences of tokens (words or subwords) using a tokenizer. These sequences are padded to a fixed length to ensure uniform input size for further analysis.

6. Splitting into Training and Testing Sets:

The dataset is divided into training and testing sets to evaluate the models' performance. A portion of the dataset is used for training the models, and the rest is reserved for testing their accuracy and generalization on unseen data.

# 3 Model


embedding_input | input: | [(None, 300)]
InputLayer | output: | [(None, 300)]
embedding | input: | (None, 300)
Embedding | output: | (None, 300, 100)
bidirectional(lstm) | input: | (None, 300, 100)
Bidirectional(LSTM) | output: | (None, 300, 200)
global_max_pooling1d | input: | (None, 300, 200)
GlobalMaxPooling1D | output: | (None, 200)
dense | input: | (None, 200)
Dense | output: | (None, 128)
dropout | input: | (None, 128)
Dropout | output: | (None, 128)
dense_1 | input: | (None, 128)
Dense | output: | (None, 64)
dropout_1 | input: | (None, 64)
Dropout | output: | (None, 64)
dense_2 | input: | (None, 64)
Dense | output: | (None, 32)
dropout_2 | input: | (None, 32)
Dropout | output: | (None, 32)
dense_3 | input: | (None, 32)
Dense | output: | (None, 1)

The model architecture is designed for sentiment analysis on the preprocessed Arabic text data. Here are the components:

1. **Embedding Layer:**

- The model begins with an Embedding layer, which transforms the input tokens (words or subwords) into dense word vectors of a fixed size (embedding dimension).

2. **Bidirectional LSTM Layer:**

- A Bidirectional LSTM (Long Short-Term Memory) layer is used to capture sequential patterns in both forward and backward directions within the text.
- This layer has 100 LSTM units with a dropout rate of 0.5 to prevent overfitting. It returns sequences for subsequent layers.

3. **Global Max Pooling Layer:**

- The GlobalMaxPooling1D layer extracts the most relevant features from the LSTM output sequences by selecting the maximum value across the sequence dimensions.

4. **Dense Layers with Dropout:**

- Three densely connected layers with 128, 64, and 32 units, respectively, are added after the pooling layer.
- Each dense layer uses ReLU activation to introduce non-linearity.
- Dropout layers with a dropout rate of 0.5 are done after each dense layer to prevent overfitting by randomly dropping a fraction of input units.

5. **Output Layer:**

- The model ends with a Dense layer containing a single unit with a sigmoid activation function.
- Sigmoid activation outputs values between 0 and 1, representing the probability of the text belonging to a particular sentiment class (binary classification).

6. **Model Compilation:**

- The model is compiled using the Adam optimizer and binary cross-entropy loss function for binary classification.
- Accuracy is chosen as the evaluation metric to measure the model's performance.

# 4 Model Training:

```
Epoch 1/4
58/58 [==============================] - 140s 2s/step - loss: 0.6938 - accuracy: 0.5012 - val_loss: 0.6909 - val_accuracy: 0.48
90
Epoch 2/4
58/58 [==============================] - 128s 2s/step - loss: 0.5921 - accuracy: 0.6859 - val_loss: 0.5182 - val_accuracy: 0.74
61
Epoch 3/4
58/58 [==============================] - 127s 2s/step - loss: 0.3661 - accuracy: 0.8790 - val_loss: 0.2348 - val_accuracy: 0.92
25
Epoch 4/4
58/58 [==============================] - 125s 2s/step - loss: 0.2028 - accuracy: 0.9511 - val_loss: 0.2399 - val_accuracy: 0.91
68
116/116 [==============================] - 17s 149ms/step - loss: 0.1145 - accuracy: 0.9646
Training Accuracy: 0.9646
50/50 [==============================] - 7s 148ms/step - loss: 0.2399 - accuracy: 0.9168
Testing Accuracy:  0.9168
```

- In the initial epoch, the model starts with a relatively low accuracy of around 50%.
- However, as the training progresses, the accuracy significantly improves with each epoch, reaching a peak training accuracy of approximately 95.11% by the end of the fourth epoch.

- The loss decreases consistently across epochs, indicating that the model is learning and making better predictions over time.
- The validation accuracy reaches approximately 92.25% by the fourth epoch, indicating the model's capability to generalize well on unseen validation data.
- The testing accuracy, which is around 91.68%, aligns closely with the validation accuracy, indicating that the model performs consistently on both validation and testing datasets.

# 5 Testing

```
Enter your statement in Arabic:  منزل رائع

1/1 [==============================] - 1s 623ms/step
The sentiment of the statement 'منزل رائع' is Positive.
```

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When the statement 'منزل رائع' was processed through the sentiment analysis model, the model predicted it as having a positive sentiment.

The model's prediction aligns with the sentiment expressed the input text, indicating that the sentiment analysis model has learned to recognize positive sentiments in Arabic text, correctly classifying 'منزل رائع' as a positive statement.